

Unit 31: Natural Logarithms

1. The idea, in plain words

You've been carefully steering around one particular antiderivative for a while now — $\int \frac{1}{x} dx$ — because the power rule breaks down exactly there (you'd be dividing by $n+1=0$). This unit finally gives you the tool that fills that gap: the natural logarithm, $\ln x$.

What is $\ln x$, really? It's the inverse function of e^x (where $e \approx 2.71828$ is a special constant that shows up throughout calculus). Just like $\sqrt{x}$ undoes squaring, $\ln x$ undoes raising e to a power. So $\ln x = k$ means exactly the same thing as $e^k = x$ — they're two ways of writing the same relationship.

Why logarithms are so useful: they turn multiplication into addition, and powers into multiplication. This isn't just a neat trick — it's a genuine simplification tool. Before differentiating a complicated expression full of products, quotients, and powers wrapped inside a $\ln(\cdot)$, it's almost always worth expanding it using log properties first — you'll often turn one nasty differentiation problem into three or four trivially easy ones.

The derivative of $\ln x$ is refreshingly simple: $\frac{d}{dx}[\ln x] = \frac{1}{x}$. And since $\ln x$ is a function like any other, the chain rule applies the moment you have $\ln(\text{something more complicated than plain } x)$.

And the antiderivative you've been waiting for:

$$\int \frac{1}{x} dx = \ln|x| + C$$

2. Toolbox

Definition (inverse relationship with e^x):

$$\ln x = k \iff e^k = x \qquad e^{\ln x} = x \qquad \ln(e^x) = x$$

Basic values: $\ln 1 = 0$, $\ln e = 1$.

Log properties (the “expand first” toolkit):

$$\ln(ab) = \ln a + \ln b \qquad \ln\left(\frac{a}{b}\right) = \ln a - \ln b \qquad \ln(a^n) = n \ln a$$

Important: these rules only work for products, quotients, and powers — never for sums. $\ln(a+b)$ does not equal $\ln a + \ln b$; there's no simplification for that at all.

Derivative:

$$\frac{d}{dx}[\ln x] = \frac{1}{x} \qquad (x > 0)$$

Chain rule version (for $\ln$ of anything more complex than plain x):

$$\frac{d}{dx}[\ln u] = \frac{u'}{u}$$

Antiderivative:

$$\int \frac{1}{x} dx = \ln|x| + C$$

Solving equations involving $\ln$: if $\ln(\text{expression}) = k$, exponentiate both sides ($e^{(\cdot)}$) to clear the log: $\text{expression} = e^k$.

3. Common mistakes

- Applying a log property to a sum. $\ln(a + b) \neq \ln a + \ln b$ — this only works for products, quotients, and powers, never addition.
- Forgetting the absolute value in the antiderivative. $\frac{1}{x}$ is defined for negative x too, even though $\ln x$ alone isn't — that's exactly why the antiderivative needs $\ln|x|$, not just $\ln x$.
- Forgetting the domain restriction $x > 0$ for $\ln x$ itself. When solving an equation like $\ln(x - 8) = 5$, remember the inside of the log must be positive — always check your solution makes sense in that light.
- Forgetting the chain rule when differentiating $\ln(u)$ for $u \neq x$. You must multiply by u' — don't just write $\frac{1}{u}$ and stop.
- Diving straight into product/quotient rule on a messy log expression instead of simplifying with log properties first. This is the single biggest time-saver in this whole unit — always check whether expanding first will make the derivative dramatically easier.

4. Problem Set

Warm-up

1. Simplify $\ln(8)$ in terms of $\ln 2$.
2. Find $\frac{d}{dx}[\ln(5x)]$.
3. Find $\frac{d}{dx}[\ln(x^2 + 1)]$.
4. Solve for x : $\ln(x) = 3$.
5. Find $\int \frac{3}{x} dx$.
6. Simplify $\ln(x^3 y^2)$ using log properties.

Standard

7. Solve for x : $\ln(x - 8) = 5$.
8. Find $\frac{d}{dx}[x \ln x]$.
9. Find $\frac{d}{dx}\left[\frac{\ln x}{x}\right]$.
10. Simplify $\ln(x^2 \sqrt{x + 3})$ using log properties, then find its derivative.
11. Find $\int \frac{2x}{x^2 + 1} dx$.
12. Solve for x : $\ln(2x + 1) = 0$.
13. Find $\frac{d}{dx}[\ln(\sin x)]$.

Challenge

14. $y = \frac{x^5}{4} \ln x - \frac{x^4}{4}$. Find $\frac{dy}{dx}$.

15. $y = \ln\left(\frac{1}{x^2\sqrt{x+1}}\right)$. Simplify first using log properties, then find $\frac{dy}{dx}$.
16. $y = \ln(3x^3e^{x^2})$. Simplify first using log properties, then find $\frac{dy}{dx}$.
17. Solve for x : $2\ln(x) + \ln(4) = \ln(36)$.
18. Find $\int \frac{x^2+1}{x} dx$.

Applied

19. A population grows according to $P(t) = 500e^{0.05t}$. Find the time t when the population reaches 1000.
20. A radioactive sample decays according to $A(t) = 100e^{-0.2t}$ (grams). Find the time t when only 25 grams remain.
21. A company's marginal cost is $MC(x) = \frac{10}{x}$ dollars per unit, for $x \geq 1$. Find the total cost increase from $x = 1$ to $x = 5$.
22. A tank's volume changes at a rate of $\frac{dV}{dt} = \frac{6}{t}$ (gallons/hour), for $t \geq 1$. Find the total volume change from $t = 1$ to $t = e$ hours.